Measure Theoretic Probability
Lesson 2
σ-Field Generated by a Collection; Borel Sets

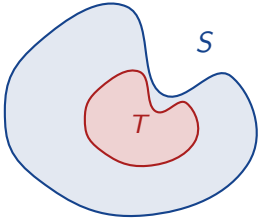
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Recap & outlook

- Ground rules for "measuring" a set.
- Recap: fields and σ-fields.
- The σ-**field generated by a collection of sets**.
- The most important σ-field on $\mathbb{R}$: the **Borel** σ-**field**.

Ground Rules for "Measuring" a Set



Agree on three rules (writing μ for the measure)

1. $\mu(S)$ is a **number**, not a letter.
2. $\mu(S) \geq 0$: a **non-negative** number.
3. If $T \subseteq S$ (one set completely inside another), then $\mu(T) \leq \mu(S)$.

A measure is a function on sets

It takes in a **set** and gives out a **number**. Example: the Lebesgue measure in 2D is the function "set $\mapsto$ area".
The sets we can measure will be called **measurable sets**; for now they are called σ-fields.

Recap: Field vs σ-Field

$\mathcal{F}$ is a collection of subsets of a non-empty set Ω

	Field	**σ-field**
(i)	$\Omega \in \mathcal{F}$	$\Omega \in \mathcal{F}$
(ii)	closed under complements	closed under complements
(iii)	closed under **finite** unions	closed under **countable** unions

"Countable" = can be indexed $1, 2, 3, \ldots$ (includes finite unions).

Last time

Every σ-field is a field, but our example $\mathcal{F}$ on $\mathbb{R}$ was a field that was **not** a σ-field: a countable union of sets in $\mathcal{F}$ gave a new set outside $\mathcal{F}$.

Notation and the σ-Field Generated by

Notation

- A: a **set** or event (e.g. "at least one tail").
- A: a **collection of sets** (a set of sets), like $\mathcal{F}$.

Lemma: The Intersection of σ-Fields Is a σ-Field

Claim

Let $\mathcal{F}_1, \mathcal{F}_2$ be σ-fields on the **same** Ω (same Ω for complements and for property (i)). Then $\mathcal{F}_1 \cap \mathcal{F}_2$ is a σ-field.

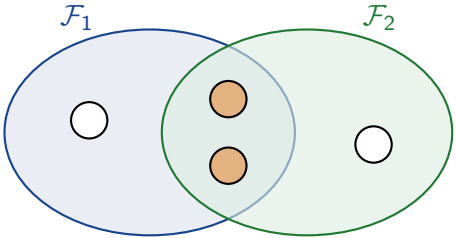
Proof

(i) $\Omega \in \mathcal{F}_1$ and $\Omega \in \mathcal{F}_2$, so $\Omega \in \mathcal{F}_1 \cap \mathcal{F}_2$.

(ii) Take $A \in \mathcal{F}_1 \cap \mathcal{F}_2$. Then $A \in \mathcal{F}_1$ and $A \in \mathcal{F}_2$, so $A^c \in \mathcal{F}_1$ and $A^c \in \mathcal{F}_2$. Hence $A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$.

(iii) Take $A_1, A_2, A_3, \cdots \in \mathcal{F}_1 \cap \mathcal{F}_2$. All lie in $\mathcal{F}_1$ and in $\mathcal{F}_2$, so $\bigcup_n A_n \in \mathcal{F}_1$ and $\bigcup_n A_n \in \mathcal{F}_2$. Hence $\bigcup_n A_n \in \mathcal{F}_1 \cap \mathcal{F}_2$. □

What $\mathcal{F}_1 \cap \mathcal{F}_2$ Contains



Each circle is a **set**; the orange ones are in both collections.

Warning

A set in $\mathcal{F}_1 \cap \mathcal{F}_2$ does **not** have the form $A_1 \cap A_2$ with $A_1 \in \mathcal{F}_1$, $A_2 \in \mathcal{F}_2$.
$\mathcal{F}_1, \mathcal{F}_2$ are **collections of sets**. A set is in the intersection if the entire set belongs to both collections. We never intersect individual sets here.

Examples 1 and 2

Example 1: one generating set

Let $= \{A\}$ (one set). Then

$$= \{\emptyset,\ A,\ A^c,\ \Omega\},$$

the σ-field generated by A from Lesson 1: throw in the complement, $\emptyset$ and Ω.

Example 2: generating from a σ-field

If $\mathcal{F}$ is already a σ-field, then

$$\mathcal{F} = \mathcal{F}.$$

We throw in nothing, since it is already a σ-field.

Example 3: The Borel σ-Field

Setting

Let $\Omega = \mathbb{R}$ and let

$$= \{ (a, b) \ : \ a < b \} \quad \text{(all open intervals)}.$$

Definition

is the famous **Borel σ-field** (or σ-algebra) on $\mathbb{R}$, denoted . Its members are called **Borel sets**.

Notation

Earlier was a generic collection of sets. From now on is reserved for the Borel sets.

Why it matters

This is the **most important σ-field in $\mathbb{R}$**. It generalizes easily to $\mathbb{R}^2, \mathbb{R}^3, \ldots$ and is central to measure theoretic probability and real analysis.

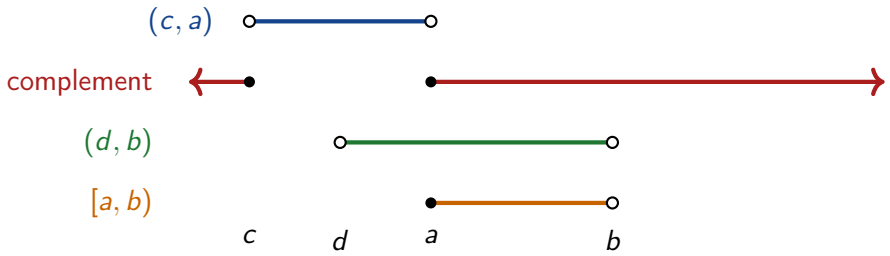
Borel Sets: Half-Open Interval $[a, b)$

Construction from open intervals

Choose numbers $c < d < a < b$. Then

$$[a, b) = \big((c, a)\big)^c \cap (d, b).$$

Take the complement of an open interval, then intersect with another open interval.



More Borel Sets

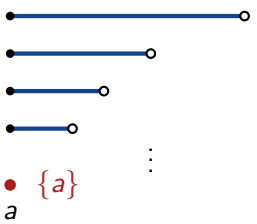
Built from the half-open intervals

- $(a, b]$: a totally symmetric argument.
- $[a, b] = [a, b) \cup (a, b]$ (closed intervals).
- $\{a\} = (c, a] \cap [a, b)$ (singletons), for $c < a < b$.

Fancier route to $\{a\}$

$$\{a\} = \bigcap_{n=1}^{\infty} \left[a,\ a + \tfrac{1}{n}\right).$$

The right endpoint keeps shrinking, the left endpoint a is always kept, and anything above a is ruled out for large n.



From Closed Intervals $[a, b]$

Direction 1: open intervals $\Rightarrow$ closed intervals

Already shown: starting from open intervals we get $[a, b]$.

Direction 2: closed intervals $\Rightarrow$ open intervals

Singletons are closed intervals: $\{a\} = [a, a]$. A σ-field is closed under complements, so $\{a\}^c$ and $\{b\}^c$ are in it. Hence

$$(a, b) = [a, b] \cap \{a\}^c \cap \{b\}^c.$$

Conclusion

Closed intervals $[a, b]$ generate **exactly the same σ-field**: the Borel sets on $\mathbb{R}$.

From Infinite Intervals $(-\infty, a]$

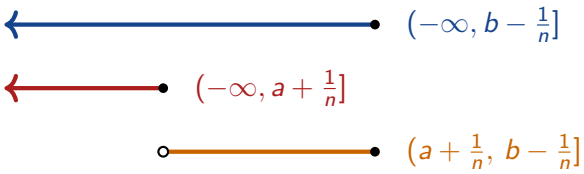
Claim

The sets $(-\infty, a]$ also generate . We must show we get (a, b) from them (Direction 1) and the converse (Direction 2).

Direction 1: step 1

For every n (large enough that $a + \frac{1}{n} < b - \frac{1}{n}$):

$$\left(a + \tfrac{1}{n},\ b - \tfrac{1}{n}\right] = \left(-\infty,\ b - \tfrac{1}{n}\right] \cap \left(-\infty,\ a + \tfrac{1}{n}\right]^c.$$



From $(-\infty, a]$: Getting (a, b)

Direction 1: step 2 (union over n)

Fix $a < b$. Start the union at an index N large enough that the first interval makes sense:

$$(a, b) = \bigcup_{n=N}^{\infty} \left(a + \tfrac{1}{n},\ b - \tfrac{1}{n}\right].$$

